

Decisive-Decision Rate (DDR): A Framework for Comparing and Calibrating Equivalence Acceptance Criteria

Abstract

When a manufacturing process undergoes a change, regulatory frameworks require demonstration that the change does not adversely affect product quality. A standard statistical approach is the Two One-Sided Test (TOST) for testing equivalence. The equivalence test requires an Equivalence Acceptance Criteria (EAC), a critical margin Δ within which pre- and post-change processes are declared equivalent. Multiple methods for choosing EAC exist in practice, yet no quantitative framework exists for empirical comparison and calibration.

In this work, we propose the **Decisive-Decision Rate (DDR) framework**, a simulation-based diagnostic based on a Monte Carlo cross-classification metric. This metric cross-classifies the joint outcome of a difference test (two-sided t -test) and an equivalence test (TOST) into four mutually exclusive zones: *concordant*-equivalent, *concordant*-different, *grey zone* (inconclusive), and *paradox* (D-test detects a difference yet TOST declares equivalence). From this 2×2 decision matrix we derive three diagnostic metrics — the DDR (proportion of concordant decisions), the Paradox Rate (PR), and the Grey Zone Rate (GZR). Defining each metric as a function of the true shift δ , we score each EAC calibration method. We further introduce the **integrated DDR (iDDR)**, a scalar summary obtained by integrating the DDR curve over a standardised shift range, which enables direct ranking of competing EAC methods.

Fundamentally, we apply the framework to compare four structurally different EAC methods — σ -scaled, percent-of-mean, effect-size-based, and a Ppk-informed margin based on process capability targets. The comparison is conducted with a factorial Monte Carlo design spanning multiple combinations of mean-to-variability ratio, spec-to-variability ratio, sample size, sample-size asymmetry, variance ratio, and process off-centring. Across a factorial grid of 4320 dimensionless parameter combinations, the σ -scaled method emerges as the most robust general-purpose choice, while the Ppk-informed method is best-calibrated at single-scenario baselines when specification limits are available. The percent-of-mean method exhibits systematically inflated paradox rates due to its disconnection from process variability.

The DDR framework is complementary to existing tools such as operating characteristic curves and second-generation p -values, and provides actionable diagnostics for margin selection in pharmaceutical manufacturing comparability studies.

1 Introduction

Pharmaceutical manufacturing processes are routinely subject to changes including site transfers, scale-ups, equipment replacements, and raw material supplier changes. Regulatory guidance FDA [FDA(2003)] and ICH Q5E [ICH(2004)] require that any such change be assessed to ensure product quality remains consistent. The standard statistical tool for this demonstration is the **Two One-Sided Test (TOST)** for equivalence [Schuirmann(1987)], which tests whether the difference between pre- and post-change means falls within a margin $\pm\Delta$. The choice of this margin—the **Equivalence Acceptance Criteria (EAC)**—is critical. A margin that is too wide risks accepting a meaningfully shifted process (false equivalence), while one that is too narrow risks rejecting a process that has not changed (false alarm).

Despite its centrality, the margin selection problem remains largely unsolved. Practitioners have proposed heuristics such as 5–15% of specification tolerance [Little(2015)], σ -scaled margins ($\Delta = \lambda \cdot \sigma_{pre}$) [Chatfield & Borman(2009), Zahel(2022)], percent-of-mean margins ($\Delta = \lambda\% |\mu_{pre}|$), effect-size-based margins grounded in hypothesis-testing theory [Burdick et al.(2017), Sec. 9.4.1], and capability-linked (Ppk-informed) margins that derive Δ from a process capability target [Burdick et al.(2017), Sec. 9.4.1]. No consensus method has emerged, and no quantitative framework exists for comparing these approaches.

Several lines of prior work address related but distinct aspects of the equivalence testing problem. [Lakens(2017)] and [Lakens et al.(2018)] proposed a four-zone classification of the joint outcome of a difference test and an equivalence test, distinguishing concordant, grey-zone, and paradox outcomes. [Isager & Fitzgerald(2024)] formalised this as “Three-Sided Testing” (TST), producing the same four-outcome partition we adopt (Table 1). However, TST remains an *interpretive procedure* for single studies; it classifies the zone of a given result but does not use zone *frequencies* across simulations as a calibration diagnostic. In contrast, we treat zone proportions as quantitative metrics for comparing and ranking margin specifications. [Blume et al.(2018)] introduced **Second-Generation P-Values (SGPV)**, which quantify CI-to-margin overlap as a continuous evidence measure, but do not compare different margin specifications. Most recently, [Koobs & Koning(2026)] proposed data-dependent margins via post-hoc *e*-value curves, solving the margin problem without pre-specification. Their approach is complementary: they address evidence reporting without a fixed margin, whereas we address margin selection when one is required by protocol.

Within pharmaceutical comparability, [Yassouridis et al.(2021)] is the closest simulation study, comparing the impact of sample sizes, data distributions, variances and EAC values on the performance of equivalence tests within a single method rather than across structurally different methods. [Chatfield & Borman(2009), Chatfield et al.(2011)] developed σ -scaled methods; [Yu et al.(2017a)] proposed a risk-based acceptance criteria that utilizing a Bayesian approach to incorporate prior scientific knowledge about process distribution; [Burdick et al.(2017)] provided a comprehensive treatment including effect-size and capability-based margins. However, the literature remains prescriptive rather than comparative: each method is advocated in isolation without quantitative benchmarks against alternatives.

In this paper, we propose the **Decisive-Decision Rate (DDR) framework**, a simulation-based diagnostic that fills this gap. The central idea is to cross-classify the joint outcome of a difference test (two-sided *t*-test) and an equivalence test (TOST) into four mutually exclusive zones—concordant-equivalent, concordant-different, grey zone (inconclusive), and paradox—then compute the *frequencies* of these zones via Monte Carlo simulation as a function of the true shift δ . From this cross-classification we derive three diagnostic curves: the DDR (proportion of concordant decisions), the Paradox Rate (PR), and the Grey Zone Rate (GZR). We further introduce the **integrated DDR (iDDR)**, a scalar summary obtained by integrating the DDR curve over a standardised shift range, which enables direct ranking of competing EAC methods.

Our methodology proceeds as follows. For each candidate EAC method and a factorial grid of dimensionless process parameters, we simulate thousands of paired datasets at each true shift value, classify each outcome into one of the four zones, and aggregate the resulting DDR, GZR, and PR curves. The iDDR scalar provides an overall ranking.

This framework makes the following contributions:

1. **Quantitative margin calibration diagnostics.** We use zone *frequencies* from Monte Carlo simulation as metrics for margin quality, providing actionable DDR, PR, and GZR diagnostics.
2. **Guaranteed decisional coherence.** A well-calibrated margin (high DDR) ensures that a difference test and equivalence test do not produce contradictory conclusions. The paradox rate explicitly quantifies this risk.
3. **Sample size and power guidance.** The framework informs minimum sample size requirements for achieving a target level of decisional coherence.
4. **A structured method-selection framework.** Systematic comparison of four EAC methods across 4320 factorial scenarios provides data-driven guidance for margin selection.

The remainder of this paper is organised as follows. Section 2 introduces notation and defines the four EAC methods. Section 3 presents the proposed DDR framework including the decision matrix, diagnostic rates, DDR curves, and iDDR. Section 4 describes the Monte Carlo simulation engine and factorial experimental design. Section 5 presents the results: baseline comparisons, sensitivity analyses, operating characteristics, DDR-based rankings, robustness analyses, and tuning-parameter optimisation. Section 6 discusses findings, practical recommendations, and limitations. Section 7 concludes.

2 Preliminaries

2.1 Notation and Assumptions

Consider two populations corresponding to a manufacturing process before and after a change. The *pre-change population* is

$$X_i \sim \mathcal{N}(\mu_{pre}, \sigma_{pre}^2), \quad i = 1, \dots, n_{pre}, \quad (1)$$

where μ_{pre} and σ_{pre} are the pre-change process mean and standard deviation. The *post-change population* is

$$Y_j \sim \mathcal{N}(\mu_{pre} + \delta_{true}, \sigma_{post}^2), \quad j = 1, \dots, n_{post}, \quad (2)$$

where δ_{true} is the *true* (unknown) mean shift caused by the process change. The product attribute is subject to a two-sided specification limits $[LSL, USL]$.

Two hypothesis tests are applied to each paired dataset:

A *difference test* (D-test), a standard two-sided *t*-test at significance level α_d to check if difference exists between pre- and post-change means:

$$H_0: \mu_{post} - \mu_{pre} = 0 \quad \text{vs} \quad H_a: \mu_{post} - \mu_{pre} \neq 0 \quad (3)$$

An *equivalence test* (E-test), the TOST procedure at level α_e , which tests whether the difference between means falls within the margin $\pm\Delta$:

$$H_0: |\mu_{post} - \mu_{pre}| \geq \Delta \quad \text{vs} \quad H_a: |\mu_{post} - \mu_{pre}| < \Delta, \quad (4)$$

where $\Delta > 0$ is the equivalence acceptance criterion (EAC).

Both tests use the pooled standard error

$$SE = S_p \sqrt{1/n_{pre} + 1/n_{post}}, \quad S_p = \sqrt{\frac{(n_{pre} - 1) S_{pre}^2 + (n_{post} - 1) S_{post}^2}{n_{pre} + n_{post} - 2}}, \quad (5)$$

with $df = n_{pre} + n_{post} - 2$ degrees of freedom. This standard error formulation assumes that pre- and post-change observations are independent, normally distributed, and equality of population variances ($\sigma_{post}/\sigma_{pre} \approx 1$).¹ In this work, we set $\alpha_d = \alpha_e = 0.05$ unless stated otherwise.

2.2 EAC Methods Under Comparison

We consider four structurally distinct methods for computing the equivalence acceptance criterion Δ . Each encodes a different philosophy about what constitutes a tolerable shift.

Method A: σ -Scaled.

$$\text{EAC}_A = \lambda \cdot \sigma_{pre} \quad (6)$$

This margin scales linearly with pre-change variability; the multiplier λ must be specified. [Chatfield & Borman(2009)] recommend $\lambda = 1.5$ in the specific context of *analytical method transfer*—comparing two laboratory assay methods for agreement. However, his heuristic ($1.5\sigma_{pre}$) was calibrated for analytical measurement variability, not for process-level comparability where shift magnitudes and noise structures can differ substantially. More broadly, guidance for λ selection is mostly domain-specific and heuristic; with no principled or application-agnostic procedure for calibrating λ , or for choosing among EAC methods in the first place. The DDR framework developed in Section 3 addresses precisely this gap.

¹In Section 5 we deliberately violate this assumption (varying $\sigma_{post}/\sigma_{pre}$ from 0.75 to 2.0) to assess the sensitivity of the DDR diagnostics to variance heterogeneity.

Method B: Percent-of-Mean.

$$\text{EAC}_B = \lambda_{\%} \cdot |\mu_{pre}| \quad (7)$$

Common in pharmaceutical QA, this margin is a fixed percentage of the process mean. Although this approach is very intuitive, it is scale-dependent: at large μ_{pre} relative to σ_{pre} , the margin becomes excessively wide and disconnected from actual process variability.

Method C: Effect-Size-Based.

$$\text{EAC}_C = ES \cdot \hat{\sigma}_{upper}, \quad ES = \frac{\delta_{assumed}}{\sigma_{pre}}, \quad \hat{\sigma}_{upper} = \sqrt{\frac{(n_{pre} - 1)s_{pre}^2}{\chi_{\alpha, n_{pre}-1}^2}} \quad (8)$$

Here $\delta_{assumed}$ is the analyst's *a priori* assumed shift magnitude (a tuning parameter, distinct from the true shift δ_{true}). The effect size $ES = \delta_{assumed}/\sigma_{pre}$ is a dimensionless quantity².

The χ^2 -corrected upper confidence bound, $\hat{\sigma}_{upper}$ protects against underestimating variability by replacing σ_{pre} with its $(1 - \alpha_{\chi^2})$ -level upper bound [Burdick et al.(2017), Sec. 9.4.1]. Here α_{χ^2} controls the conservatism of the variance correction³. Additionally, the χ^2 variance correction is sensitive to sample size so that as n_{pre} increases, the multiplier approaches 1⁴.

The effect size also acts as a direct multiplier, the margin is explicitly tied to the analyst's assumptions about the true shift magnitude.

Remark 1 (Relationship to Method A). *Setting $c = \sqrt{(n_{pre} - 1)/\chi_{\alpha, n_{pre}-1}^2} \geq 1$,*

$$\text{EAC}_C = \underbrace{ES}_{\lambda} \cdot \underbrace{\sigma_{pre} \cdot c}_{\hat{\sigma}_{upper}} \xrightarrow{n_{pre} \rightarrow \infty} \underbrace{ES}_{\lambda} \cdot \sigma_{pre} = \text{EAC}_A.$$

The only conceptual difference is that Method C inflates σ_{pre} by c to account for estimation uncertainty from finite data, whereas Method A takes σ_{pre} at face value.

Although this method is grounded in hypothesis-testing theory, both sources of uncertainty—the subjective choice of ES and the statistical correction for variance—compound multiplicatively.

Method D: Ppk-Informed. Unlike the preceding methods which define the margin from summary statistics of the pre-change distribution alone, Method D anchors the margin to the process capability via the process performance index Ppk [Burdick et al.(2017), Sec. 9.4.1]. Given a target capability Ppk_{target} that must be maintained after a process change, the largest mean shift the process can absorb without violating that target defines the margin. For a centred process this is

$$\text{EAC}_D = \max\left(\frac{USL - LSL}{2} - 3 \cdot Ppk_{target} \cdot \sigma_{pre}, 0\right). \quad (9)$$

We derive this result formally below, beginning with the standard definition of Ppk , and then generalise it to the off-centre case (Corollary 1), which is the form adopted throughout the paper.

Definition 1 (Process Performance Index). *For a process with mean μ and standard deviation σ operating within specification limits $[LSL, USL]$, the process performance index Ppk is defined as*

$$Ppk = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right). \quad (10)$$

² $ES = 0.5$ corresponds to a shift of half a standard deviation, regardless of the absolute scale of the data.

³e.g. $\alpha_{\chi^2} = 0.01$ inflate $\hat{\sigma}_{upper}$, widening the margin

⁴At $n_{pre} = 10$, $\alpha_{\chi^2} = 0.05$, the multiplier is $\approx 1.6\times$, whereas at $n_{pre} = 30$, the multiplier is closer to $1.2\times$

Proposition 1 (Ppk-Informed EAC — Centred Case). *Consider a centred pre-change process with mean μ_{pre} , standard deviation σ_{pre} , and specification limits $[LSL, USL]$.*

Say the process undergoes a mean shift δ after a process change, with unchanged variance, $\sigma_{post} = \sigma_{pre}$. Then the largest tolerable shift for which $Ppk_{post} \geq Ppk_{target}$ —which defines the margin—is

$$EAC_D = \max\left(\frac{USL - LSL}{2} - 3 \cdot Ppk_{target} \cdot \sigma_{pre}, 0\right). \quad (11)$$

Proof. Let $\mu_{pre} = \frac{USL+LSL}{2}$ be the pre-change process mean. From equation (10) we have that

$$Ppk_{pre} = \min\left(\frac{USL - \mu_{pre}}{3\sigma_{pre}}, \frac{\mu_{pre} - LSL}{3\sigma_{pre}}\right) \quad (12)$$

$$= \min\left(\frac{USL - \frac{USL+LSL}{2}}{3\sigma_{pre}}, \frac{\frac{USL+LSL}{2} - LSL}{3\sigma_{pre}}\right) \quad (13)$$

$$= \min\left(\frac{\frac{USL-LSL}{2}}{3\sigma_{pre}}, \frac{\frac{USL-LSL}{2}}{3\sigma_{pre}}\right) \quad (14)$$

$$= \min\left(\frac{S}{3\sigma_{pre}}, \frac{S}{3\sigma_{pre}}\right) = \frac{S}{3\sigma_{pre}}, \quad (15)$$

where $S = \frac{USL-LSL}{2}$ is the half-width of the specification interval.

After a mean shift δ , ($\mu_{post} = \mu_{pre} + \delta$), from equation (15), the post-change process capability index is:

$$Ppk_{post} = \min\left(\frac{S - \delta}{3\sigma_{pre}}, \frac{S + \delta}{3\sigma_{pre}}\right) = \frac{S - |\delta|}{3\sigma_{pre}}. \quad (16)$$

Imposing the capability constraint $Ppk_{post} \geq Ppk_{target}$:

$$\frac{S - |\delta|}{3\sigma_{pre}} \geq Ppk_{target} \implies |\delta| \leq S - 3 \cdot Ppk_{target} \cdot \sigma_{pre}. \quad (17)$$

From equation (17), the maximum tolerable shift is $S - 3 \cdot Ppk_{target} \cdot \sigma_{pre}$.

Thus when $3 \cdot Ppk_{target} \cdot \sigma_{pre} > S$, the capability target is infeasible even at zero shift; clamping at zero and substituting $S = \frac{USL-LSL}{2}$ yields (11). $\square$

Corollary 1 (Off-Centre Correction). *When $\mu_{pre} \neq \frac{USL+LSL}{2}$, the process is closer to one specification limit than the other, so the effective margin is no longer S but the distance d_{pre} from μ_{pre} to the nearest specification limit,*

$$d_{pre} = \min(USL - \mu_{pre}, \mu_{pre} - LSL) \leq S.$$

The Ppk-informed EAC that guarantees $Ppk_{post} \geq Ppk_{target}$ regardless of shift direction is then

$$EAC_D = \max(d_{pre} - 3 Ppk_{target} \sigma_{pre}, 0). \quad (18)$$

This robust form reduces to (11) when the process is centred ($d_{pre} = S$), and is the operative definition of EAC_D in the remainder of the paper.

Proof. Replace S by d_{pre} in the proof of Proposition 1; the algebra is identical. Since $d_{pre} \leq S$, the centred formula (11) overestimates the allowable shift when the process is off-centre. $\square$

Remark 2 (Feasibility Boundary and Capability Retention). *The robust margin (18) decreases linearly in Ppk_{target} and vanishes once*

$$Ppk_{target} \geq \frac{d_{pre}}{3\sigma_{pre}} = Ppk_{pre}, \quad (19)$$

beyond which no shift is tolerable (Figure 1); the ceiling $S/(3\sigma_{pre})$ is reached only by a perfectly centred process. Anchoring directly to the current capability ($Ppk_{target} = Ppk_{pre}$) thus collapses the margin to zero. To retain a usable margin we scale by a capability retention fraction $\rho \in (0, 1)$, setting $Ppk_{target} = \rho Ppk_{pre}$, whereupon σ_{pre} cancels:

$$EAC_D = \max(d_{pre} - 3\rho Ppk_{pre} \sigma_{pre}, 0) = d_{pre} (1 - \rho). \quad (20)$$

All experiments adopt $\rho = 0.8$ (retain 80% of pre-change capability) unless stated otherwise; a fixed target such as $Ppk_{target} = 1.33$ may be used instead when a regulatory threshold applies.

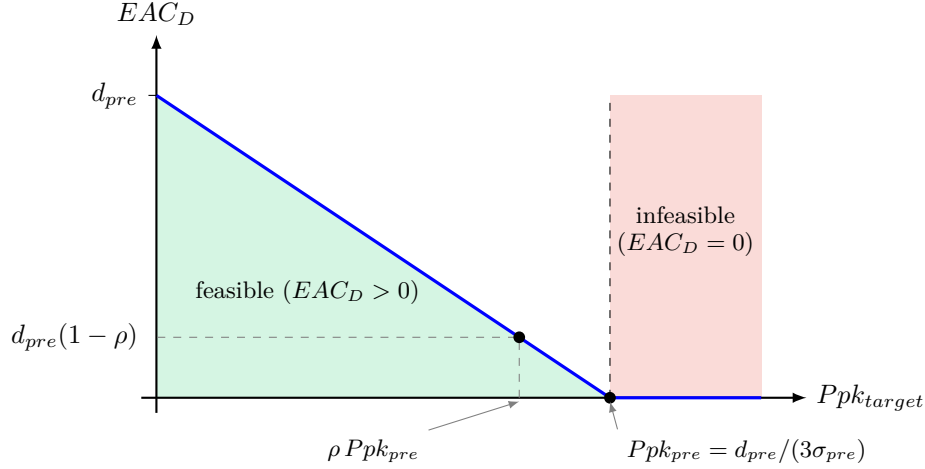


Figure 1: Feasibility boundary for the Ppk-informed margin. EAC_D decreases linearly with the target capability Ppk_{target} and vanishes once Ppk_{target} reaches the pre-change capability $Ppk_{pre} = d_{pre}/(3\sigma_{pre})$. The capability retention fraction ρ selects an operating point inside the feasible region, yielding $EAC_D = d_{pre}(1 - \rho)$ (shown for $\rho = 0.8$).

3 The Decisive-Decision Rate (DDR) Framework

The DDR framework is *method-agnostic*. Given any candidate margin, Δ (however derived), it returns a quantitative diagnostic of that margin's calibration quality.

3.1 Decision Matrix and Diagnostic Rates

When both a difference test (two-sided t -test at α_d) and an equivalence test (TOST at α_e) are applied to the same data, their joint outcome falls into one of four mutually exclusive zones. We define the 2×2 decision matrix as follows:

From this classification we define three diagnostic rates:

$$DDR = (Z_1 + Z_4)/N \quad (\text{proportion of concordant decisions}) \quad (21)$$

$$PR = Z_3/N \quad (\text{paradox rate: margin too wide}) \quad (22)$$

$$GZR = Z_2/N \quad (\text{grey zone rate: margin too narrow or underpowered}) \quad (23)$$

By construction, $DDR + PR + GZR = 1$ for every shift value. A high DDR indicates a well-calibrated margin; elevated PR signals that the margin is too wide (concludes equivalence despite a detected difference); elevated GZR signals that the margin is too narrow or insufficiently powered for a decisive outcome.

	E-test pass	E-test fail
D-test fail	$Z_1 \checkmark$ Concordant-equivalent	$Z_2 ?$ Grey zone
D-test pass	$Z_3 !$ Paradox	$Z_4 \checkmark$ Concordant-different

Table 1: Decision matrix notation. Z_1 – Z_4 denote zone counts from N Monte Carlo replicates.

3.2 The DDR Curve: $DDR(\delta)$

For a given EAC method and parameter setting, the DDR curve $DDR(\delta/\sigma_{pre})$ traces the proportion of concordant decisions as a function of the standardised true shift δ/σ_{pre} . Its shape follows directly from the joint behaviour of the two tests:

- *Small shifts* ($\delta \ll \text{EAC}$): the D-test detects no difference and TOST confirms equivalence, so the tests agree and $DDR \approx 1$.
- *Large shifts* ($\delta \gg \text{EAC}$): the D-test detects the difference and TOST rejects equivalence, so the tests again agree and $DDR \approx 1$.
- *Near the boundary* ($\delta \approx \text{EAC}$): the shift lies close to both decision thresholds at once, the tests frequently disagree, and the curve dips.

The dip is the diagnostic signature of the method; its depth, width, and location are interpreted against the simulated curves in Section 5.4.

3.3 The Integrated DDR (iDDR)

$$iDDR = \frac{1}{\delta_{max}/\sigma} \int_0^{\delta_{max}/\sigma} DDR(\delta/\sigma) d(\delta/\sigma) \quad (24)$$

A scalar summary for ranking methods. Higher iDDR indicates better overall margin calibration across the operating range. The iDDR collapses the full DDR curve into a single number, enabling direct comparison and ranking of competing EAC methods.

4 Experimental Setup

4.1 Populations and Data-Generating Process

Each Monte Carlo replicate draws independent samples from two normal populations:

$$X_i \sim \mathcal{N}(\mu_{pre}, \sigma_{pre}^2), \quad i = 1, \dots, n_{pre} \quad (\text{pre-change}), \quad (25)$$

$$Y_j \sim \mathcal{N}(\mu_{pre} + \delta_{true}, \sigma_{post}^2), \quad j = 1, \dots, n_{post} \quad (\text{post-change}), \quad (26)$$

where δ_{true} is the true mean shift imposed by the simulation. The EAC for each method is computed from the pre-change parameters $(\mu_{pre}, \sigma_{pre}, n_{pre})$ and specification limits *before* generating post-change data; the post-change population parameters $(\delta_{true}, \sigma_{post}, n_{post})$ affect only the test statistics. This mirrors practice: the margin is set prospectively from historical data, then applied to a new validation batch.

4.2 Monte Carlo Simulation Engine

For each EAC method $M \in \{A, B, C, D\}$ and each standardised shift $\delta_{true}/\sigma_{pre}$, the simulation proceeds as follows.

1. Draw N_{sim} independent pre-change samples of size n_{pre} from (25) and N_{sim} independent post-change samples of size n_{post} from (26).
2. For each replicate $i = 1, \dots, N_{sim}$, compute the sample means $\bar{X}_{pre}^{(i)}, \bar{Y}_{post}^{(i)}$, the sample standard deviations $S_{pre}^{(i)}, S_{post}^{(i)}$, the pooled standard error $SE^{(i)}$, and the mean difference $D^{(i)} = \bar{Y}_{post}^{(i)} - \bar{X}_{pre}^{(i)}$.
3. Apply the D-test: reject H_0 if $|D^{(i)}|/SE^{(i)} > t_{1-\alpha_d/2, df}$, where $df = n_{pre} + n_{post} - 2$.
4. Apply the E-test (TOST): declare equivalence if both $(D^{(i)} + \Delta_M)/SE^{(i)} > t_{1-\alpha_e, df}$ and $(\Delta_M - D^{(i)})/SE^{(i)} > t_{1-\alpha_e, df}$, where Δ_M is the EAC for method M .
5. Classify each replicate into zones Z_1 – Z_4 and compute DDR, GZR, PR .

All operations are vectorised over the N_{sim} dimension using NumPy broadcasting, eliminating Python-level loops over replicates. The pooled (equal-variance) t -test is used throughout, with $df = n_{pre} + n_{post} - 2$.⁵

4.3 Dimensionless Parameterisation

As shown in Section 2, all four EAC methods produce margins that depend on the pre-change parameters only through dimensionless ratios. Moreover, the TOST and D-test statistics are invariant to the scale of the data: under the transformation $X \mapsto X/\sigma_{pre}$, both numerator and denominator of the t -statistic scale by $1/\sigma_{pre}$ and cancel. Consequently, all test decisions—and hence DDR, PR, GZR —depend on the EAC method only through the standardised margin Δ/σ_{pre} , the standardised true shift $\delta_{true}/\sigma_{pre}$, the sample sizes n_{pre} and n_{post} , and the variance ratio $\sigma_{post}/\sigma_{pre}$.

This observation motivates a factorial grid expressed entirely in dimensionless quantities, eliminating redundant runs that would differ only in absolute scale. The five dimensionless drivers are:

1. μ_{pre}/σ_{pre} — the mean-to-variability ratio (affects Method B only, via $EAC_B/\sigma_{pre} = \lambda_{\%} \cdot |\mu_{pre}/\sigma_{pre}|$);
2. S/σ_{pre} — the spec-to-variability ratio, where $S = (USL - LSL)/2$ (affects Method D, via $EAC_D/\sigma_{pre} = \max(d_{pre}/\sigma_{pre} - 3 Ppk_{target}, 0)$);
3. n_{pre} — the pre-change sample size (affects EAC computation for Method C and test power);
4. n_{post}/n_{pre} — the sample-size ratio (affects test power but not the EAC);
5. $\sigma_{post}/\sigma_{pre}$ — the variance ratio (affects test power and robustness).

An additional parameter specific to Method D is the off-centre fraction η , defined below.

4.4 Factorial Design

Off-Centre Process Mean. In practice, a manufacturing process is never perfectly centred between its specification limits. To reflect this, we parameterise the process mean as

$$\mu_{pre} = \frac{USL + LSL}{2} + \eta \cdot S, \quad (27)$$

where $\eta \in (-1, 1)$ is the *off-centre fraction*: $\eta = 0$ is perfectly centred, $|\eta| = 1$ places μ_{pre} at a specification limit. For Method D, we set $Ppk_{target} = \rho \cdot Ppk_{pre}$ with $\rho = 0.8$, retaining 80% of the pre-change capability (see Remark 2):

$$Ppk_{target} = \rho \cdot Ppk_{pre} = \frac{\rho \cdot d_{pre}}{3 \sigma_{pre}} = \frac{\rho \cdot S(1 - |\eta|)}{3 \sigma_{pre}}. \quad (28)$$

⁵The pooled t -test assumes $\sigma_{pre} = \sigma_{post}$. We retain it even when $\sigma_{post}/\sigma_{pre} \neq 1$ in order to assess the robustness of the DDR diagnostics under a deliberate violation of this assumption; see Section 5.

Sample-Size Constraints. Pre-change sample sizes n_{pre} range from 5 to 30 in increments of 5, reflecting the small batch sizes typical in pharmaceutical manufacturing. The post-change sample size is set as $n_{post} = \lfloor r \cdot n_{pre} \rfloor$ for ratio $r \in \{1/3, 1/2, 1, 2\}$, subject to the constraint $5 \leq n_{post} \leq 30$. This design captures the practical asymmetry where the pre-change dataset may be historical (larger) or the post-change validation batch may be limited (smaller), or vice versa.

Factorial Grid. Table 2 defines the full factorial grid. All parameters except n_{pre} and n_{post} are dimensionless. For the simulation, we fix $\sigma_{pre} = 1$ without loss of generality (since all test decisions depend only on the dimensionless ratios above) and derive the absolute parameters as $\mu_{pre} = (\mu_{pre}/\sigma_{pre}) \cdot \sigma_{pre}$, $USL - LSL = 2(S/\sigma_{pre})\sigma_{pre}$, centred at $\mu_{pre} - \eta S$.

Table 2: Factorial grid: dimensionless process parameters. All quantities except n_{pre} are ratios; $\sigma_{pre} = 1$ is fixed without loss of generality.

Parameter	Symbol	Grid values	Count
Mean-to-variability ratio	μ_{pre}/σ_{pre}	5, 10, 25, 50	4
Spec-to-variability ratio	S/σ_{pre}	2.5, 5, 10	3
Pre-change sample size	n_{pre}	5, 10, 15, 20, 25, 30	6
Sample-size ratio	n_{post}/n_{pre}	$\frac{1}{3}, \frac{1}{2}, 1, 2$	4
Variance ratio	$\sigma_{post}/\sigma_{pre}$	0.75, 1.0, 1.25, 1.5, 2.0	5
Off-centre fraction	η	0.05, 0.10, 0.20, 0.35	4

This yields $4 \times 3 \times 6 \times 4 \times 5 \times 4 = 5,760$ parameter combinations. After applying the sample-size constraint $5 \leq n_{post} \leq 30$, some combinations are excluded (e.g. $n_{pre} = 5$ with $r = 1/3$ gives $n_{post} \approx 2$, which is dropped). For each valid combination, the DDR, GZR, and PR are computed at 20 equally spaced standardised shifts $\delta_{true}/\sigma_{pre} \in [0, 3.0]$ using $N_{sim} = 5,000$ Monte Carlo replicates per shift.

Tuning-Parameter Adaptation. For each factorial combination, the EAC tuning parameters are adapted to the scenario rather than held at fixed baseline values:

- Method A: $\lambda = 1.5$ (the recommendation of [Chatfield & Borman(2009)]);
- Method B: $\lambda_{\%} = 0.05$ (a moderate 5% margin);
- Method C: $ES = 0.5$ (a “small” assumed effect), $\alpha_{\chi^2} = 0.05$;
- Method D: $Ppk_{target} = 0.8 Ppk_{pre}$ (retain 80% of pre-change capability; see Remark 2).

Tuning-Parameter Optimisation. To identify the best-case performance of each method, we sweep its primary tuning parameter over a fine grid while holding process parameters at their factorial values:

- Method A: $\lambda \in \{0.5, 0.75, 1.0, 1.25, 1.5, 1.75, 2.0, 2.5, 3.0\}$ (9 values);
- Method B: $\lambda_{\%} \in \{0.01, 0.02, 0.03, 0.05, 0.08, 0.10, 0.15, 0.20\}$ (8 values);
- Method C: $ES \in \{0.1, 0.2, 0.3, 0.5, 0.75, 1.0, 1.5, 2.0\}$ (8 values);
- Method D: $Ppk_{target}/Ppk_{pre} \in \{0.5, 0.67, 0.8, 0.9, 1.0, 1.1, 1.2\}$ (7 values).

For each tuning value, the full factorial grid is run at 12 standardised shifts with $N_{sim} = 2,000$. The optimal tuning parameter θ^* for each method is defined as $\theta^* = \arg \max_{\theta} \overline{iDDR}(\theta)$, where $\overline{iDDR}$ is the iDDR averaged over all factorial combinations. This sweep separates the question “which method is inherently better?” from “are some methods being unfairly penalised by suboptimal tuning?”

4.5 Baseline Parameterisation

The baseline scenario uses $\mu_{pre}/\sigma_{pre} = 50$, $S/\sigma_{pre} = 5$, $n_{pre} = n_{post} = 20$, $\sigma_{post}/\sigma_{pre} = 1.0$, and $\eta = 0.05$ (slight off-centre). With $\sigma_{pre} = 1$, this gives $\mu_{pre} \approx 50.25$ (slightly off-centre), $S = 5$, $USL = 55$, $LSL = 45$, and $d_{pre} = 4.75$. The tuning parameters are: $\lambda = 1.5$ (Method A), $\lambda_{\%} = 0.05$ (Method B), $ES = 0.5$ with $\alpha_{\chi^2} = 0.05$ (Method C), and $Ppk_{target} = 0.8$ $Ppk_{pre} \approx 1.27$ (Method D). Under these settings the four methods produce the following standardised margins:

Method	EAC/ σ_{pre}	Source of margin	Key driver
A (σ -scaled)	1.50	λ	Tuning parameter
B (%-of-mean)	2.51	$\lambda_{\%} \cdot \mu_{pre}/\sigma_{pre}$	μ_{pre}/σ_{pre} ratio
C (effect-size)	0.69	$ES \cdot \hat{\sigma}_{upper}/\sigma_{pre}$	ES , n_{pre}
D (Ppk-informed)	0.95	$d_{pre}(1 - \rho)/\sigma_{pre}$	d_{pre}/σ_{pre} , η , ρ

5 Results

Sections 5.1–5.4 present results evaluated at the fixed baseline parameterisation (Section 4.5): $n_{pre} = n_{post} = 20$, $\mu_{pre}/\sigma_{pre} = 50$, $S/\sigma_{pre} = 5$, $\eta = 0.05$, $\sigma_{post}/\sigma_{pre} = 1$. Sections 5.1–5.3 are purely analytical—EAC values are computed directly from their defining formulae and operating characteristic curves from the exact non-central t distribution—while Section 5.4 applies the DDR framework via Monte Carlo simulation at the same baseline parameters. Because all test decisions depend only on dimensionless ratios (Section 4.3), these results generalise to any absolute scale: a process with $\sigma_{pre} = 10$ and $\mu_{pre} = 500$ yields identical standardised margins and power curves.

5.1 Baseline EAC Comparison

EAC values are computed directly from each method’s defining formula at the baseline parameters (no simulation).

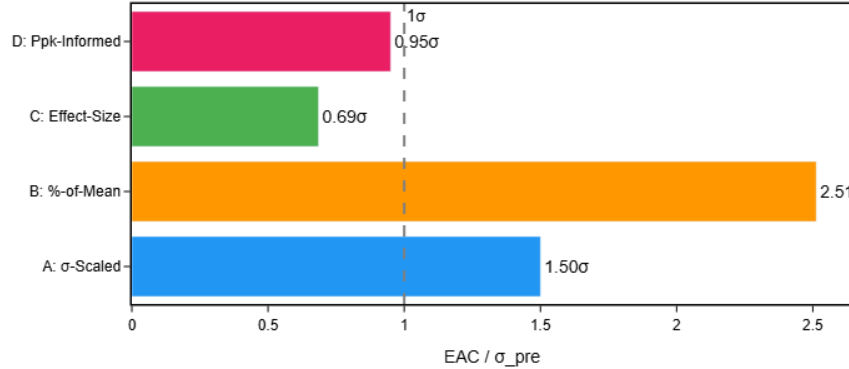


Figure 2: Baseline EAC values (standardised by σ_{pre}) for the four methods, computed analytically at the baseline parameterisation ($\mu/\sigma = 50$, $S/\sigma = 5$, $n_{pre} = n_{post} = 20$, $\eta = 0.05$). Dashed line marks the $1\sigma_{pre}$ reference.

Figure 2 presents the four baseline EAC values as a forest plot on the standardised EAC/σ_{pre} axis. The ordering from tightest to widest is $C < D < A < B$. Method C produces the tightest margin ($0.69\sigma_{pre}$), falling below the $1\sigma_{pre}$ reference; at the baseline sample size $n = 20$ per group this margin does not achieve 80% TOST power at $\delta_{true} = 0$ (Table 3), highlighting the conservatism inherent in the χ^2 variance correction at moderate sample sizes. Method D is also below $1\sigma_{pre}$ ($0.95\sigma_{pre}$) but reaches 80% power at $n = 15$, making it practical at the baseline sample size. Method A ($1.50\sigma_{pre}$) sits above the $1\sigma_{pre}$ line and requires only $n = 7$ for 80% power, offering a comfortable balance between TOST power and decisional coherence.

Method B's wide margin ($2.51 \sigma_{pre}$, driven by the large μ_{pre}/σ_{pre} ratio) is wide enough that TOST will declare equivalence for shifts that the D-test can readily detect, foreshadowing elevated paradox rates.

5.2 Sensitivity Analysis

Each method's primary tuning parameter is swept independently while all other parameters are held at their baseline values (no simulation; EAC values computed analytically). Figure 3 shows the resulting sensitivity curves.

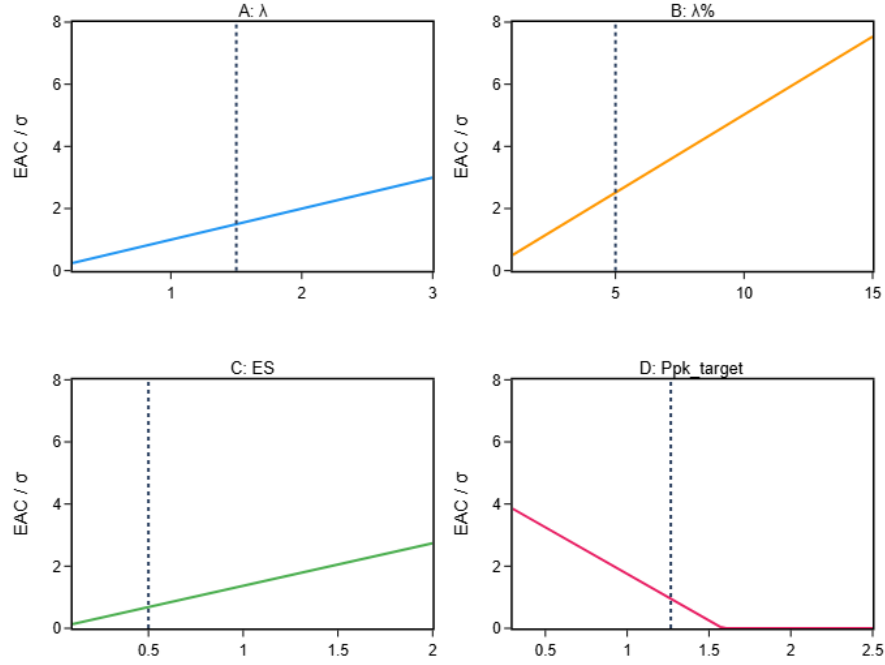


Figure 3: Analytical EAC sensitivity to each method's primary tuning parameter at the baseline parameterisation ($\mu/\sigma = 50$, $S/\sigma = 5$, $n_{pre} = n_{post} = 20$, $\eta = 0.05$). Panels share a common y -axis; dotted lines mark baseline tuning values ($\lambda = 1.5$, $\lambda\% = 0.05$, $ES = 0.5$, $\rho = 0.8$).

Method A (λ). The EAC is linear in λ , with slope σ_{pre} . Over the practical range $\lambda \in [0.5, 3.0]$, the margin spans $[1.0, 6.0]$. The response is predictable and there are no critical thresholds, making Method A straightforward to tune.

Method B ($\lambda\%$). The EAC is also linear in $\lambda\%$ but its magnitude depends on μ_{pre}/σ_{pre} . At $\mu_{pre}/\sigma_{pre} = 50$, a 5% margin ($\lambda\% = 0.05$) gives $EAC/\sigma_{pre} = 2.5$. The scale dependence is evident: to achieve a $1\sigma_{pre}$ margin at $\mu_{pre}/\sigma_{pre} = 50$ requires $\lambda\% = 0.02$ (2%), a value rarely used in practice.

Method C (ES). The EAC is super-linear in ES because the effect size enters multiplicatively with the χ^2 -inflated $\hat{\sigma}_{upper}$. Small changes in ES produce large changes in the margin, especially at small n_{pre} where the χ^2 inflation is most pronounced. This sensitivity is a practical concern: the analyst must specify ES *a priori*, and mis-specifications are amplified.

Method D (Ppk_{target}). The EAC decreases linearly in Ppk_{target} and hits a floor at zero when $Ppk_{target} \geq Ppk_{pre} = d_{pre}/(3\sigma_{pre})$. For the baseline scenario, the critical value is $Ppk_{target} = Ppk_{pre}$; beyond this point

no shift is tolerable (see Remark 2). The linear structure means that the sensitivity is uniform across the operating range, with no abrupt cliffs.

Overall, Methods A and D are the most robust to parameter mis-specification (linear, moderate slope), while Method C is the most sensitive (steep, non-linear near the baseline).

5.3 Operating Characteristics

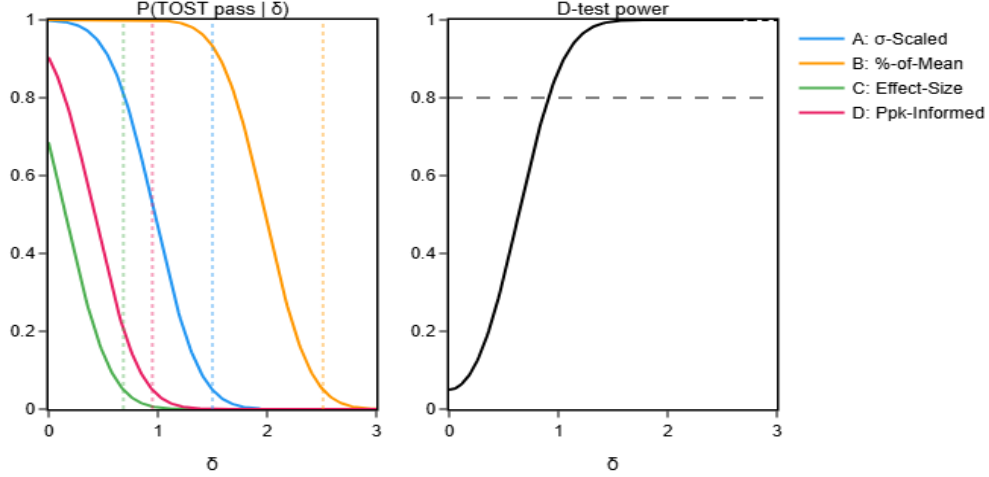


Figure 4: Analytical operating characteristic curves at the baseline parameterisation ($n_{pre} = n_{post} = 20$, $\sigma_{post}/\sigma_{pre} = 1$), computed via the exact non-central t distribution. Left: TOST pass probability vs. true shift δ ; dotted lines mark each method’s EAC. Right: D-test power (common to all methods); dashed grey line marks 80% power.

All curves are computed analytically from the exact non-central t distribution at the baseline parameterisation (no Monte Carlo simulation). Figure 4 presents the operating characteristic curves for all four methods. For each method M with margin Δ_M , the TOST pass probability at a true shift δ is

$$P(\text{TOST pass} \mid \delta) = \min(F_{t'}(-t_{1-\alpha_e, df}; \lambda_{nc}^{(U)}), 1 - F_{t'}(t_{1-\alpha_e, df}; \lambda_{nc}^{(L)})), \quad (29)$$

where $F_{t'}(\cdot; \lambda)$ is the non-central t CDF with non-centrality parameters $\lambda_{nc}^{(U)} = (\delta_{true} - \Delta_M)/\sigma_{eff}$ and $\lambda_{nc}^{(L)} = (\delta_{true} + \Delta_M)/\sigma_{eff}$, and $\sigma_{eff} = \sqrt{\sigma_{pre}^2/n_{pre} + \sigma_{post}^2/n_{post}}$. The D-test power curve is obtained analogously from the two-sided rejection probability of the non-central t distribution with $\lambda_{nc}^{(D)} = \delta_{true}/\sigma_{eff}$.

The TOST curves confirm that each method is properly calibrated: $P(\text{TOST pass} \mid \delta_{true} = \Delta_M) \leq \alpha_e$ for all four methods. Method B, with the widest margin, retains high TOST power over a broad shift range, while Method D’s tight margin causes TOST power to drop rapidly. Notably, Method C does not achieve 80% TOST power even at $\delta_{true} = 0$: at the baseline sample size $n = 20$, $P(\text{TOST pass} \mid \delta = 0) \approx 0.70$ (see also Table 3, which shows $n = 28$ is required for 80% power). This highlights the inherent conservatism of the χ^2 variance correction at moderate sample sizes—the inflated $\hat{\sigma}_{upper}$ produces a margin too tight for TOST to reliably confirm equivalence without larger samples. The D-test power curve is common to all methods (since it does not depend on the EAC); its power depends on n_{pre} , n_{post} , and the variance ratio.

Using the same non-central t machinery (known- σ assumption), Table 3 reports the minimum equal-group sample size required to achieve 80% and 90% TOST power at $\delta_{true} = 0$ under the baseline parameterisation. Appendix C validates these values against Monte Carlo simulation with estimated variance, showing that the analytical values are optimistic lower bounds. The required n scales inversely with $(\text{EAC}/\sigma_{pre})^2$: Method C requires $n = 28$ for 80% power (explaining its under-powered OC curve above), while Method B needs only $n = 3$. This power–precision trade-off is a fundamental consideration when selecting an EAC method.

Table 3: Minimum sample size per group ($n_{pre} = n_{post}$) for target TOST power at $\delta_{true} = 0$, computed analytically via exact non-central t power at the baseline parameterisation ($\mu/\sigma = 50$, $S/\sigma = 5$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$). See Appendix C for MC validation.

Method	EAC/ σ_{pre}	n for 80% power	n for 90% power
A (σ -scaled)	1.50	7	9
B (%-of-mean)	2.51	3	4
C (effect-size)	0.69	28	38
D (Ppk-informed)	0.95	15	20

5.4 Baseline DDR Diagnostic Profile

Having characterised each method’s margin and power analytically, we now apply the DDR framework via Monte Carlo simulation at the same baseline parameters. For each method, $N_{sim} = 5,000$ replicates are generated at $N_\delta = 20$ equally spaced shifts $\delta/\sigma_{pre} \in [0, 3]$, and each replicate is classified into zones Z_1 – Z_4 .

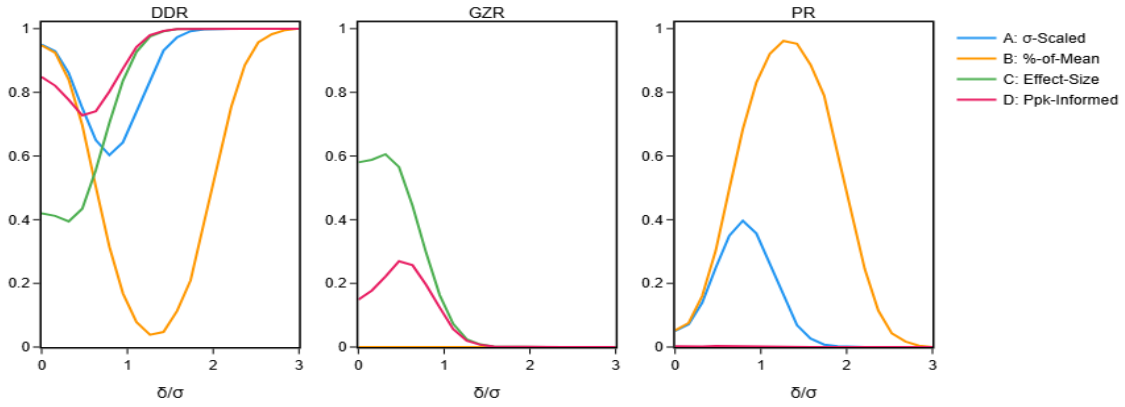


Figure 5: Monte Carlo DDR, GZR, and PR curves at the baseline parameterisation ($\mu/\sigma = 50$, $S/\sigma = 5$, $n_{pre} = n_{post} = 20$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$; $N_{sim} = 5,000$, $N_\delta = 20$ shifts over $[0, 3\sigma_{pre}]$). Left: DDR exhibits a U-shape with dips tracking each method’s EAC. Centre: GZR peaks near the EAC boundary. Right: PR shows Method B with high paradox rates; Methods C and D near zero.

All four DDR curves display the characteristic U-shape: $DDR \approx 1$ at both $\delta = 0$ and $\delta/\sigma_{pre} \geq 2.5$, with a dip near each method’s EAC boundary whose location tracks the ordering $C < D < A < B$. The morphology of this dip is itself diagnostic: a shallow, narrow dip indicates a well-calibrated margin with a sharp transition from equivalence to difference, whereas a deep, wide dip marks a region of the shift axis over which the two tests frequently contradict one another. Tight-margin methods (C, D) produce the highest grey-zone rates; Method B produces the highest paradox rate, peaking near $\delta/\sigma_{pre} \approx 1.5$ – 2.0 . Methods C and D have near-zero paradox rates, confirming that tight margins eliminate paradoxes at the cost of elevated inconclusiveness.

Table 4 summarises the diagnostic performance via iDDR and worst-case rates. Method D achieves the highest iDDR (0.925) with near-zero paradox rate (0.003). Method A ranks second (0.889); Method C eliminates paradoxes entirely but has the highest grey-zone peak (0.606). Method B ranks last (0.550) with a paradox rate of 0.962.

This baseline diagnostic profile establishes the single-scenario behaviour; Sections 5.5–5.7 generalise these patterns across the full factorial grid.

Table 4: Baseline diagnostic summary from Monte Carlo simulation ($\mu/\sigma = 50$, $S/\sigma = 5$, $n_{pre} = n_{post} = 20$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$; $N_{sim} = 5,000$, $N_\delta = 20$). iDDR is the area-normalised integral of the DDR curve over $\delta/\sigma_{pre} \in [0, 3]$; remaining columns report extrema. Methods ranked by iDDR (descending).

Method	EAC/ σ_{pre}	iDDR	min DDR	max GZR	max PR
D: Ppk-Informed	0.95	0.925	0.727	0.269	0.003
A: σ -Scaled	1.50	0.889	0.603	0.000	0.397
C: Effect-Size	0.69	0.839	0.394	0.606	0.000
B: %-of-Mean	2.51	0.550	0.038	0.000	0.962

5.5 Method Ranking

Sections 5.5–5.7 use Monte Carlo simulation across the full factorial grid (Section 4.4): 4320 parameter combinations $\times$ 4 methods, with $N_{sim} = 1,500$ replicates per shift point and $N_\delta = 12$ shift values over $[0, 3\sigma_{pre}]$. Table 5 ranks the four methods by six diagnostic criteria averaged over this grid: iDDR, iGZR, iPR (integrated rates), the worst-case minimum DDR and maximum GZR and PR, and mean EAC/ σ_{pre} .

Table 5: Method ranking via Monte Carlo simulation across the full factorial grid (4320 scenarios $\times$ 4 methods; $N_{sim} = 1,500$, $N_\delta = 12$ shifts over $[0, 3\sigma_{pre}]$). iDDR, iGZR, and iPR are area-normalised integrals; remaining columns report averages of per-scenario extrema. Ranked by iDDR (descending).

Method	iDDR	$\pm$ std	iDDR _{min}	iGZR	iPR	max GZR	max PR	EAC/ σ
A: σ -Scaled	0.843	0.128	0.345	0.119	0.039	0.222	0.134	1.50
D: Ppk-Informed	0.747	0.137	0.316	0.230	0.023	0.605	0.069	0.96
B: %-of-Mean	0.720	0.135	0.316	0.204	0.077	0.549	0.167	1.18
C: Effect-Size	0.705	0.119	0.323	0.295	<0.001	0.799	<0.001	0.76

Method A achieves the highest mean iDDR (0.843) across the factorial grid, demonstrating the greatest robustness. Method D ranks second (0.747), trading some concordance for a near-zero paradox rate. No single method dominates all criteria: Method C eliminates paradoxes entirely (max PR < 0.001) but at the cost of the highest grey-zone rates (max GZR = 0.799), while Method B produces the highest paradox rates.

Figure 6 shows the full iDDR distribution across all scenarios. Method A has the highest median and tightest spread; Method C has the lowest median but narrowest range; Methods B and D show wider distributions with longer lower tails.

5.6 Robustness Analysis

The following robustness analyses are derived from the same factorial MC simulation described above, marginalising the iDDR results over all factors except the one under investigation.

5.6.1 Sample Size Sensitivity

Figure 7 plots iDDR against n_{pre} , averaged over all remaining factorial parameters. All methods improve monotonically with n_{pre} ; Method C converges most slowly due to its χ^2 -inflated variance correction. The ANOVA analysis (Appendix B) confirms that sample size is the single largest driver of iDDR ($\eta^2 = 0.218$). However, this aggregate improvement masks a competing effect: as n grows, the difference test becomes increasingly overpowered, detecting arbitrarily small—and practically meaningless—shifts. Because the equivalence margin Δ is fixed, a statistically significant but negligible difference can fall inside the equivalence region, inflating the paradox rate (Z_3). This phenomenon is most pronounced for wider-margin methods (B, A) and represents a fundamental tension between the point-null D-test and large-sample regimes.

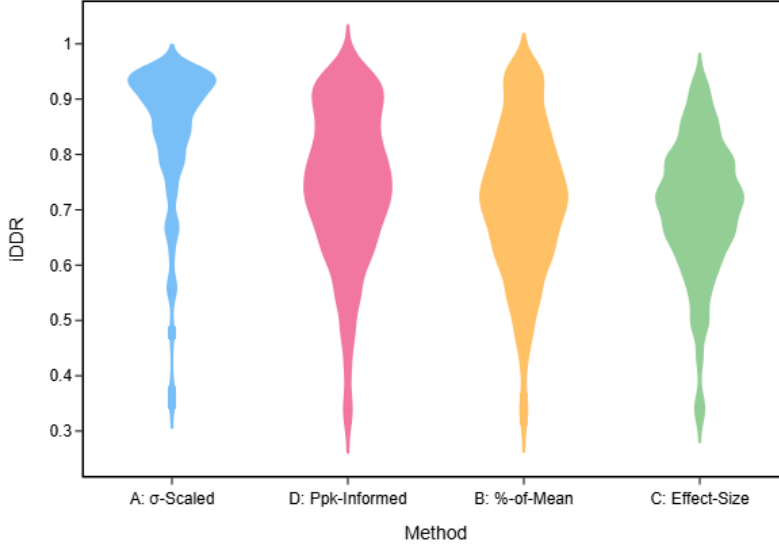


Figure 6: Violin plots of iDDR across all 4 320 factorial scenarios (Monte Carlo, $N_{sim} = 1,500$, $N_{\delta} = 12$). Box shows the interquartile range; horizontal line marks the mean.

5.6.2 Unequal Variance

Figure 8 shows iDDR under five variance ratios, averaged over all remaining factorial parameters. All methods degrade as $\sigma_{post}/\sigma_{pre}$ increases, with Methods C and D most affected (their margins depend on σ_{pre} alone). Method A is the most robust to variance inflation. The ANOVA (Appendix B) ranks variance ratio as the third-largest factor ($\eta^2 = 0.122$).

5.7 Tuning-Parameter Optimisation

For each method, we sweep its primary tuning parameter and, at every candidate value, run the full Monte Carlo simulation engine (Section 3) across a subset of factorial scenarios ($n_{pre} \in \{10, 20, 30\}$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$; 36 $\mu/\sigma \times S/\sigma$ combinations per n_{pre} level) to obtain the mean iDDR. Each reported iDDR is therefore averaged over thousands of MC replicates across dozens of design configurations, not a single-scenario calculation. A coarse sweep identifies approximate optima, after which a fine grid of 15 values centred on each θ^* refines the estimate. Figure 9 shows the fine-tuned iDDR curves with the optimising values $\lambda^* \approx 1.36$ (A), $\lambda_{\%}^* \approx 0.031$ (B), $ES^* \approx 0.95$ (C), and $\rho^* \approx 0.83$ (D).

A natural question is whether, once each method is optimally tuned, their peak performances converge. They do not. Even at their respective optima, peak iDDR values remain well separated: Method C achieves 0.916, Method A 0.900, Method D 0.806, and Method B 0.789 (Table 6). The ≈ 13 percentage-point gap between the best (C) and worst (B) performers persists because tuning can only optimise *within* each method’s structural constraints—it cannot overcome a method’s fundamental sensitivity to the design factors that vary across the factorial grid. Method C’s χ^2 -corrected margin adapts naturally to sample size, while Method B’s dependence on the process mean level creates an irreducible handicap.

6 Discussion

6.1 Summary of Findings

1. **Method A (σ -Scaled)** achieves the highest mean iDDR across the full factorial grid (0.843) and the tightest distribution, making it the most robust general-purpose choice. Its performance is stable across sample sizes, variance ratios, and specification widths.

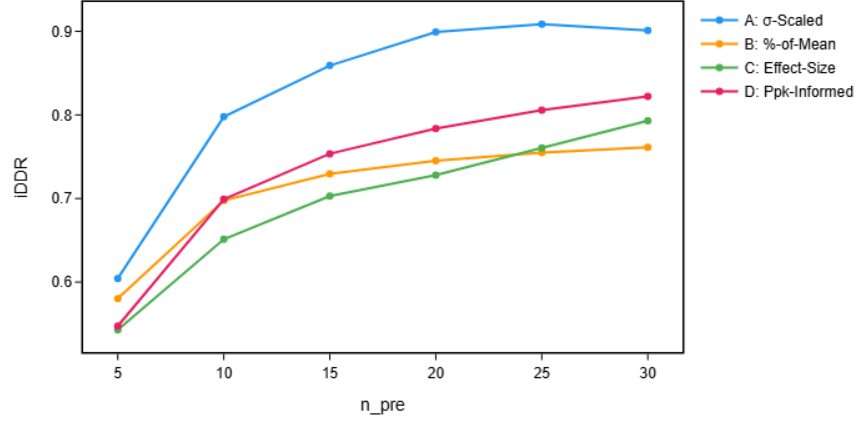


Figure 7: Mean iDDR vs. n_{pre} from the factorial Monte Carlo simulation ($N_{sim} = 1,500$, $N_{\delta} = 12$), marginalised over all remaining design factors (μ/σ , S/σ , n_{post}/n_{pre} , $\sigma_{post}/\sigma_{pre}$, η).

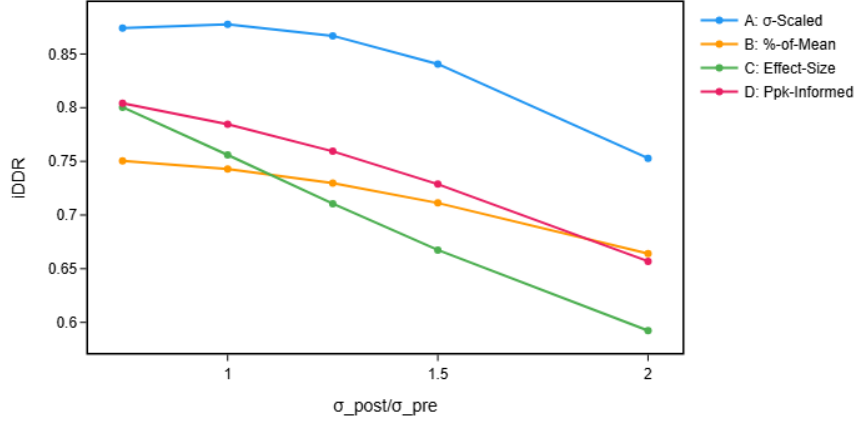


Figure 8: Mean iDDR vs. variance ratio $\sigma_{post}/\sigma_{pre}$ from the factorial Monte Carlo simulation ($N_{sim} = 1,500$, $N_{\delta} = 12$), marginalised over all remaining design factors.

2. **Method D (Ppk-Informed)** ranks second overall (iDDR = 0.747) and is best-calibrated at the baseline parameterisation (iDDR = 0.925). It provides the most scientifically justified margin when specification limits are available, with near-zero paradox rates, but is more sensitive to design conditions than Method A.
3. **Method C (Effect-Size)** is the most conservative method, virtually eliminating paradox risk (max PR < 0.001) at the cost of the highest grey-zone rates (max GZR = 0.799) and largest sample-size requirements. When optimally tuned ($ES^* \approx 0.95$), it achieves the highest peak iDDR of any method (0.916, Table 6), though this advantage does not transfer to the default factorial ranking. Best for safety-critical applications.
4. **Method B (Percent-of-Mean)** ranks third in iDDR but produces the highest paradox rates across the factorial grid. Its margin depends on the process mean level, which the ANOVA confirms has negligible effect on the other three methods ($\eta^2 < 0.001$). Not recommended as a standalone method.

Notably, fine-tuning each method's primary parameter to its iDDR-maximising value does *not* lead to convergent performance (Section 5.7). Even at their respective optima, a ≈ 13 percentage-point gap persists

Table 6: Fine-tuned optima from Monte Carlo tuning sweep ($n_{pre} \in \{10, 20, 30\}$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$; 36 $\mu/\sigma \times S/\sigma$ combinations per n_{pre} ; fine grid of 15 values per method). Peak iDDR at each method’s optimal tuning parameter θ^* .

Method	Parameter	θ^*	Peak iDDR
C: Effect-Size	ES	0.95	0.916
A: σ -Scaled	λ	1.36	0.900
D: Ppk-Informed	ρ	0.83	0.806
B: %-of-Mean	$\lambda_{\%}$	0.031	0.789

between the best- and worst-performing methods, confirming that the ranking differences are structural rather than artefacts of default tuning choices.

6.2 Decision Framework

Based on the results above, we propose the following decision framework for practitioners selecting an EAC method.

1. Are specification limits available?

- *Yes*: Proceed to step 2.
- *No*: Use Method A (σ -scaled) with $\lambda \in [1.0, 1.5]$, or Method C (effect-size) for high-sensitivity attributes.

2. Is the spec-to-variability ratio S/σ wide (≥ 4) or narrow (< 4)?

- *Wide* ($S/\sigma \geq 4$): Method D (Ppk-informed) with $Ppk_{target} = 1.33$ is recommended. The margin will be well calibrated and the required sample size manageable.
- *Narrow* ($S/\sigma < 4$): Method D produces a very tight margin (near zero for $Ppk_{target} \geq Ppk_{pre}$), requiring impractically large samples. Use Method A with a conservative $\lambda \leq 1.5$, or Method C for safety-critical attributes.

3. Is the application routine or high-sensitivity?

- *Routine*: Method A provides the best balance of simplicity, interpretability, and robustness.
- *High-sensitivity* (e.g. *potency*, *immunogenicity*): Method C minimises paradox risk at the cost of elevated grey zone rates and larger sample sizes.

In all cases, Method B (percent-of-mean) should be avoided as a standalone method due to its systematic dependence on the process mean level.

6.3 Relationship to Existing Methods

- **SGPV** [Blume et al.(2018)]: Complementary. SGPV measures CI-to-margin overlap within a single framework; the DDR framework cross-classifies two separate test decisions.
- **Three-Sided Testing** [Isager & Fitzgerald(2024)]: The DDR framework uses the same four-outcome classification but as a simulation-based calibration tool rather than a single-study interpretive guide.
- **Data-Dependent Margins** [Kooobs & Koning(2026)]: Solves the margin problem via post-hoc e -value curves; complementary to the pre-specified margin comparison addressed here.

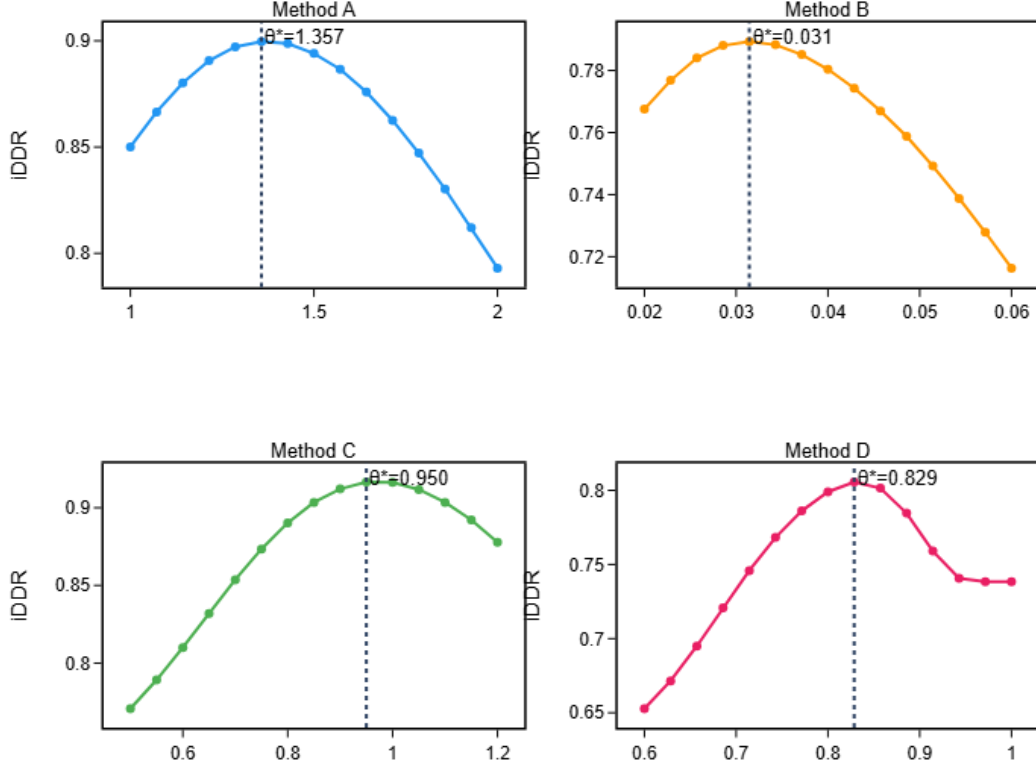


Figure 9: Fine-tuned iDDR vs. primary tuning parameter, evaluated via Monte Carlo simulation over a factorial subset ($n_{pre} \in \{10, 20, 30\}$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$; 36 $\mu/\sigma \times S/\sigma$ combinations). Dotted lines mark the iDDR-maximising value θ^* .

6.4 Generality of the DDR Framework

The DDR framework is agnostic to both the EAC method and the application domain. Any candidate margin—Bayesian, data-dependent [KooBs & Koning(2026)], domain-specific, or entirely novel—can be evaluated by computing its DDR, PR, and GZR curves. Rather than advocating a particular formula, the framework provides a *calibration diagnostic* for justifying margin choices with quantitative evidence, applicable to process comparability, analytical method transfer, site transfer, or scale-up.

6.5 Limitations

- All methods assume normality and known/estimated σ_{pre} .
- Method D requires specification limits; its margin depends on the off-centre fraction η and the choice of Ppk_{target} .
- The iDDR integrates over a fixed shift range $[0, 3\sigma_{pre}]$ — sensitivity to this range should be assessed.
- Simulation results depend on the chosen factorial grid; the sample-size range ($n_{pre} \in [5, 30]$) reflects pharmaceutical practice but may not generalise to other domains.
- At large sample sizes the point-null D-test becomes overpowered, flagging trivially small shifts as statistically significant and inflating the paradox rate. The DDR framework as formulated uses a classical point-null test, and does not incorporate a practical-significance gate on the difference test.

6.6 Future Work

- **Bayesian EAC:** Incorporate prior distributions on shift magnitude.
- **Adaptive methods:** Re-estimate σ from incoming post-change data.
- **Non-normal extensions:** Extend the DDR framework to skewed or heavy-tailed distributions.
- **Multivariate CQAs:** Extend to simultaneous equivalence testing across multiple quality attributes.
- **Practical-significance gatekeeping:** Replace the point-null D-test with a minimum-effect gate—for example, requiring the observed standardised effect size (Cohen’s d) to exceed a clinically meaningful threshold before a difference is declared. Such a gate would suppress spurious paradoxes at large n while preserving the DDR framework’s zone classification.

7 Conclusion

The Decisive-Decision Rate framework provides a principled, quantitative tool for comparing and calibrating Equivalence Acceptance Criteria. By cross-classifying the joint outcome of a difference test and an equivalence test across thousands of simulated scenarios, the DDR, PR, and GZR metrics reveal whether a given margin is too wide (high PR), too narrow (high GZR), or well-calibrated (high DDR). The integrated DDR enables direct ranking of competing methods.

Applied to four structurally different EAC methods across a factorial grid of 4 320 dimensionless parameter combinations, the framework identifies the σ -scaled approach (Method A) as the most robust general-purpose choice, achieving the highest mean iDDR (0.843) with the tightest distribution across design conditions. The Ppk-informed approach (Method D) is the best-calibrated at the baseline parameterisation (iDDR = 0.925) and provides the strongest scientific justification when specification limits are available. The effect-size approach (Method C) is the safest for high-sensitivity applications, eliminating paradox risk entirely. The percent-of-mean method (Method B) is consistently outperformed due to its dependence on the process mean level. Importantly, fine-tuning each method to its optimal tuning parameter does not eliminate these ranking differences: peak iDDR values range from 0.916 (C) to 0.789 (B), confirming that the performance gaps are structural.

An ANOVA of the factorial results (Appendix B) reveals that sample size ($\eta^2 = 0.218$), method choice ($\eta^2 = 0.145$), and variance ratio ($\eta^2 = 0.122$) are the dominant drivers of diagnostic concordance, while the process mean level and off-centre fraction have negligible effect sizes—confirming the scale-agnosticism of the dimensionless parameterisation.

We recommend that practitioners in pharmaceutical manufacturing adopt the DDR framework as a standard diagnostic when selecting or validating EAC methods, alongside traditional operating characteristic analyses.

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A Metric-Space Properties of the DDR Framework

We prove that the diagnostic quantities of Section 3 induce well-defined metrics and pseudo-metrics (in the sense of real analysis) on the space of EAC methods.

A.1 Preliminaries

Definition 2 (Metric). *A function $d: X \times X \rightarrow \mathbb{R}$ is a metric on a set X if, for all $x, y, z \in X$:*

- (M1) $d(x, y) \geq 0$ (non-negativity)
- (M2) $d(x, y) = 0 \iff x = y$ (identity of indiscernibles)
- (M3) $d(x, y) = d(y, x)$ (symmetry)
- (M4) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality)

If only (M1), (M3), (M4) hold and (M2) is weakened to $d(x, x) = 0$, then d is a pseudo-metric.

A.2 Setup and Notation

Let $\mathcal{M}$ denote the set of EAC methods under comparison. For each $M \in \mathcal{M}$, the Monte Carlo simulation produces three diagnostic curves on the standardised-shift domain $I = [0, \delta_{\max}/\sigma]$:

$$DDR_M, GZR_M, PR_M: I \rightarrow [0, 1], \quad DDR_M(t) + GZR_M(t) + PR_M(t) = 1 \quad \forall t \in I.$$

These bounded functions on a compact interval belong to $L^p(I)$ for every $1 \leq p \leq \infty$.

The integrated DDR is

$$iDDR(M) = \frac{1}{|I|} \int_I DDR_M(t) dt \in [0, 1]. \quad (30)$$

We define two equivalence relations:

- $M_1 \sim M_2 \iff DDR_{M_1}(t) = DDR_{M_2}(t) \text{ a.e. on } I;$
- $M_1 \approx M_2 \iff DDR_{M_1} = DDR_{M_2}, GZR_{M_1} = GZR_{M_2}, PR_{M_1} = PR_{M_2} \text{ a.e.}$

A.3 The L^p -DDR Distance

Definition 3 (L^p -DDR distance). For $1 \leq p < \infty$, define

$$d_p(M_1, M_2) = \left(\frac{1}{|I|} \int_I |DDR_{M_1}(t) - DDR_{M_2}(t)|^p dt \right)^{1/p}. \quad (31)$$

For $p = \infty$, define $d_\infty(M_1, M_2) = \text{ess sup}_{t \in I} |DDR_{M_1}(t) - DDR_{M_2}(t)|$.

Theorem 1 (d_p is a metric on $\mathcal{M}/\sim$). For every $1 \leq p \leq \infty$, the function d_p is a pseudo-metric on $\mathcal{M}$ and a metric on $\mathcal{M}/\sim$.

Proof. Define $\varphi: \mathcal{M} \rightarrow L^p(I)$, $M \mapsto DDR_M$. Then $d_p(M_1, M_2) = |I|^{-1/p} \|\varphi(M_1) - \varphi(M_2)\|_{L^p}$ (with the convention $|I|^{-1/\infty} = 1$). Scaling a norm by a positive constant yields a norm, so the axioms reduce to those of the L^p norm.

(M1). $\|\cdot\|_{L^p} \geq 0$ implies $d_p \geq 0$.

(M3). $|a - b| = |b - a|$ pointwise, so the integral is symmetric.

(M4). By Minkowski's inequality:

$$\begin{aligned} d_p(M_1, M_3) &= |I|^{-1/p} \|\varphi(M_1) - \varphi(M_3)\|_{L^p} \\ &\leq |I|^{-1/p} (\|\varphi(M_1) - \varphi(M_2)\|_{L^p} + \|\varphi(M_2) - \varphi(M_3)\|_{L^p}) \\ &= d_p(M_1, M_2) + d_p(M_2, M_3). \end{aligned}$$

(M2). $d_p(M_1, M_2) = 0 \Leftrightarrow DDR_{M_1} = DDR_{M_2}$ a.e., i.e. $M_1 \sim M_2$. On $\mathcal{M}/\sim$ this becomes $[M_1] = [M_2]$. $\square$

Remark 3. On $\mathcal{M}$ itself (without quotienting), d_p is a pseudo-metric. In practice, the four methods studied produce distinct DDR curves, so d_p is effectively a metric.

A.4 The iDDR Dissimilarity

Definition 4 (iDDR distance).

$$d_{\text{iDDR}}(M_1, M_2) = |iDDR(M_1) - iDDR(M_2)|. \quad (32)$$

Proposition 2 (d_{iDDR} is a pseudo-metric dominated by d_1).

1. d_{iDDR} is a pseudo-metric on $\mathcal{M}$.
2. $d_{\text{iDDR}}(M_1, M_2) \leq d_1(M_1, M_2)$ for all M_1, M_2 .

Proof. Part 1. The map $\psi: M \mapsto iDDR(M) \in \mathbb{R}$ composes with the absolute value, which is a metric on $\mathbb{R}$. The pull-back of a metric through any map satisfies (M1), (M3), (M4) and $d(M, M) = 0$, hence is a pseudo-metric. Full (M2) can fail: methods with different DDR curves but equal integrals give $d_{\text{iDDR}} = 0$.

Part 2. By Jensen's inequality applied to the convex function $|\cdot|$ and the probability measure $\frac{1}{|I|} dt$:

$$d_{\text{iDDR}}(M_1, M_2) = \left| \frac{1}{|I|} \int_I (DDR_{M_1} - DDR_{M_2}) dt \right| \leq \frac{1}{|I|} \int_I |DDR_{M_1} - DDR_{M_2}| dt = d_1(M_1, M_2). \quad \square$$

A.5 The Diagnostic-Profile Distance

The L^p -DDR distance uses only the DDR curve. We now define a finer distance that incorporates all three diagnostic curves, distinguishing methods that agree on DDR but decompose failures differently.

Definition 5 (Diagnostic-profile distance).

$$d_{\text{prof}}(M_1, M_2) = \frac{1}{|I|} \int_I |DDR_{M_1}(t) - DDR_{M_2}(t)| + |GZR_{M_1}(t) - GZR_{M_2}(t)| + |PR_{M_1}(t) - PR_{M_2}(t)| dt. \quad (33)$$

We first establish a pointwise identity that is central to the analysis.

Lemma 1 (Sum-zero absolute-value identity). *If $a + b + c = 0$ for $a, b, c \in \mathbb{R}$, then*

$$|a| + |b| + |c| = 2 \max(|a|, |b|, |c|). \quad (34)$$

Proof. Without loss of generality let $|a| = \max(|a|, |b|, |c|)$. From $a = -(b + c)$ and $|a| \geq |b|, |c|$ it follows that b and c cannot share the same sign as a (otherwise $|b + c| \leq |b| + |c|$ with both summands having the same sign as a gives $|a| = |b + c| = |b| + |c|$ only if both share the sign, but then $|a| = |b| + |c|$, and since $|a| \geq |b|, |c|$, we already reach the conclusion). More directly: $b + c = -a$ with $|b| \leq |a|$ and $|c| \leq |a|$ forces b and c to have the opposite sign of a (or be zero). Hence $|b| + |c| = |b + c| = |a|$, giving $|a| + |b| + |c| = 2|a|$. $\square$

Theorem 2 (d_{prof} is a pseudo-metric; a metric on $\mathcal{M}/\approx$).

1. d_{prof} is a pseudo-metric on $\mathcal{M}$.
2. $d_{\text{prof}}(M_1, M_2) = 0 \Leftrightarrow M_1 \approx M_2$ (all three curves agree a.e.).
3. Hence d_{prof} is a metric on $\mathcal{M}/\approx$.

Proof. Part 1. Write $d_{\text{prof}} = d_1^{\text{DDR}} + d_1^{\text{GZR}} + d_1^{\text{PR}}$, where each summand is the normalised L^1 distance for one diagnostic curve. Each is a pseudo-metric (by Theorem 1 applied to each curve individually), and a non-negative linear combination of pseudo-metrics is again a pseudo-metric.

Part 2. $d_{\text{prof}} = 0$ iff all three integrands vanish a.e., which is $M_1 \approx M_2$.

Part 3. Parts 1 and 2 together yield (M1)–(M4) on the quotient $\mathcal{M}/\approx$. $\square$

Proposition 3 (d_1 and d_{prof} : one-sided bound). *For all $M_1, M_2 \in \mathcal{M}$:*

$$2 d_1(M_1, M_2) \leq d_{\text{prof}}(M_1, M_2). \quad (35)$$

Equality holds when $|\Delta D| = \max(|\Delta D|, |\Delta G|, |\Delta P|)$ a.e. No finite reverse bound $d_{\text{prof}} \leq C \cdot d_1$ exists.

Proof. Write $\Delta D = \text{DDR}_{M_1} - \text{DDR}_{M_2}$, $\Delta G = \text{GZR}_{M_1} - \text{GZR}_{M_2}$, $\Delta P = \text{PR}_{M_1} - \text{PR}_{M_2}$. The unit-sum constraint gives $\Delta D + \Delta G + \Delta P = 0$ pointwise. By Lemma 1, $|\Delta D| + |\Delta G| + |\Delta P| = 2 \max(|\Delta D|, |\Delta G|, |\Delta P|) \geq 2|\Delta D|$. Integrating yields $d_{\text{prof}} \geq 2 d_1$. Equality holds, e.g., when $\Delta G = -\Delta D$ and $\Delta P = 0$.

No reverse bound exists: one can have $\Delta D(t) \approx 0$ while $\Delta G(t) = -\Delta P(t) \gg 0$ (two methods with nearly identical DDR but different GZR/PR decompositions), giving $d_1 \approx 0$ but $d_{\text{prof}} > 0$. This is precisely the added discriminative power of d_{prof} . $\square$

A.6 Summary of the Metric Hierarchy

Combining the results above, the four distances are ordered by refinement for any $M_1, M_2 \in \mathcal{M}$:

$$d_{\text{iDDR}}(M_1, M_2) \leq d_1(M_1, M_2) \leq \frac{1}{2} d_{\text{prof}}(M_1, M_2), \quad (36)$$

and $d_1(M_1, M_2) \leq d_{\infty}(M_1, M_2)$ (since $\|f\|_{L^1} \leq |I| \|f\|_{L^\infty}$ after normalisation).

Distance	Metric on	Pseudo-metric on	Information used
d_{iDDR}	$\mathcal{M}/\ker \psi$	$\mathcal{M}$	Scalar $i\text{DDR}$ only
d_p ($1 \leq p \leq \infty$)	$\mathcal{M}/\sim$	$\mathcal{M}$	DDR curve
d_{prof}	$\mathcal{M}/\approx$	$\mathcal{M}$	DDR + GZR + PR curves

Table 7: Summary of metric-space structures induced by the DDR framework. $\ker \psi = \{(M_1, M_2) : i\text{DDR}(M_1) = i\text{DDR}(M_2)\}$.

The hierarchy shows that d_{iDDR} is the coarsest comparator (only the integral matters), d_p captures the full shape of the DDR curve, and d_{prof} is the finest, distinguishing methods that agree on DDR but differ in their failure-mode profiles.

B ANOVA of Factorial Design Factors

To identify which design factors most influence the diagnostic performance of EAC methods, we fit a Type II ANOVA model to the factorial results of Section 5.5:

$$\text{iDDR} = \beta_0 + \text{Method} + \sum_j \beta_j x_j + \sum_j \gamma_j (\text{Method} \times x_j) + \varepsilon,$$

where the design factors x_j are n_{pre} , $\sigma_{post}/\sigma_{pre}$, S/σ_{pre} , μ/σ_{pre} , and η . The effect size is quantified by $\eta^2 = SS_{\text{factor}}/SS_{\text{total}}$, and interpreted following Cohen’s benchmarks: small ($\eta^2 < 0.01$), medium (0.01–0.06), and large ($\eta^2 > 0.14$). Table 8 reports all terms with $p < 0.05$.

Table 8: Type II ANOVA for iDDR from the factorial Monte Carlo simulation (17 280 observations = 4 320 scenarios $\times$ 4 methods; $N_{sim} = 1,500$, $N_\delta = 12$). All terms shown are significant at $p < 0.05$. $\eta^2 = SS_{\text{factor}}/SS_{\text{total}}$.

Factor	SS	F	p	η^2	Interpretation
n_{pre}	73.93	8 435	< 0.0001	0.218	Large
Method	49.36	1 877	< 0.0001	0.145	Large
$\sigma_{post}/\sigma_{pre}$	41.53	4 738	< 0.0001	0.122	Large
Method $\times$ S/σ_{pre}	12.68	482	< 0.0001	0.037	Medium
S/σ_{pre}	4.17	476	< 0.0001	0.012	Medium
Method $\times$ $\sigma_{post}/\sigma_{pre}$	3.95	150	< 0.0001	0.012	Medium
Method $\times$ n_{pre}	2.59	98	< 0.0001	0.008	Small
Method $\times$ η	0.18	7	0.0001	< 0.001	Negligible
η	0.06	7	0.007	< 0.001	Negligible
Method $\times$ μ/σ_{pre}	0.14	5	0.001	< 0.001	Negligible
μ/σ_{pre}	0.05	5	0.021	< 0.001	Negligible
Residual	151.24	—	—	0.445	—

Three factors account for nearly 50% of total variance:

1. **Sample size** ($\eta^2 = 0.218$): The single largest driver. All methods improve with larger n , but at different rates—Method C converges most slowly due to its χ^2 -corrected variance inflation.
2. **Method choice** ($\eta^2 = 0.145$): Confirms that the choice of EAC formula has a substantive effect on diagnostic concordance, justifying the need for systematic comparison via the DDR framework.
3. **Variance ratio** ($\eta^2 = 0.122$): Post-change variance inflation degrades all methods, but Methods A and D are more robust than B and C.

Conversely, the process mean level (μ/σ_{pre}) and off-centre fraction (η) have negligible effect sizes ($\eta^2 < 0.001$), confirming the scale-agnosticism of the dimensionless parameterisation (Section 4.3). The specification width (S/σ_{pre}) has a moderate main effect but interacts strongly with Method ($\eta^2 = 0.037$), reflecting the fact that Method D’s margin depends directly on specification limits while Method A’s does not.

C Monte Carlo Validation of Sample-Size Requirements

Table 3 in Section 5.3 reports the minimum sample size for target TOST power using the exact non-central t distribution, which assumes the population variance σ_{pre}^2 is known. In practice, σ_{pre}^2 is estimated from the data via the pooled sample variance s_p^2 . To assess the impact of variance estimation, we repeat the sample-size determination using Monte Carlo simulation: for each candidate n (with $n_{pre} = n_{post} = n$), we generate $N_{sim} = 20,000$ replicate pairs at $\delta_{true} = 0$ and $\sigma_{post}/\sigma_{pre} = 1$, compute the TOST decision using the estimated pooled standard error, and record the empirical pass rate. The minimum n at which this pass rate exceeds the target power defines the MC-based sample-size requirement.

Table 9: Minimum sample size per group at the baseline parameterisation ($\mu/\sigma = 50$, $S/\sigma = 5$, $\sigma_{post}/\sigma_{pre} = 1$, $\eta = 0.05$): analytical (non-central t , known σ) versus Monte Carlo simulation (estimated σ , $N_{sim} = 20,000$, $\delta_{true} = 0$).

Method	80% power		90% power	
	Analytical	MC	Analytical	MC
A (σ -scaled)	7	9	9	11
B (%-of-mean)	3	4	4	5
C (effect-size)	28	38	38	47
D (Ppk-informed)	15	20	20	25

The MC-based sample sizes are consistently larger than the analytical values, reflecting the additional uncertainty from estimating σ_{pre} from finite samples. The discrepancy is most pronounced for Method C (28 vs. 38 at 80% power), whose tight margin amplifies the effect of variance estimation noise. For Method B, the gap is only one sample (3 vs. 4) because its wide margin provides a large buffer against estimation error. These results confirm that the analytical values in Table 3 represent optimistic lower bounds; practitioners should use the MC-based values for realistic sample-size planning.